mRehab: A Clinically Grounded Smartphone System for Home Rehabilitation

A phone inside 3D-printed household objects counts and scores arm exercises at home

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Why home practice is hard

After a stroke, survivors need intensive, repeated arm practice. Once supervised therapy ends, they and their caregivers must keep the exercises going and make sure they are done correctly.

mRehab uses an ordinary smartphone inside a 3D-printed household object to count and score each exercise at home.



Figure A: *The kit: a smartphone, a printed mug and bowl, a storage box, and a floor mat.*

Try the demo

A mirrored display shows the phone's screen as you try the printed mug.

1. Browse the activity library. Say "slow pour with mug", set a repetition goal, and start.
2. Confirm the selected activity and count on screen. A short countdown follows.
3. Secure the phone inside the mug and pour toward the bowl.
4. Watch completed repetitions count up. Follow the next-movement cues and visual, spoken, and pouring-sound feedback.
5. Open the performance and progress views to connect your movement to its count, time, and score.
6. Repeat the sequence in airplane mode.

Pick an exercise by voice

Voice can pick the activity, set the goal, and start practice while the phone is inside the object. Touch controls stay available.

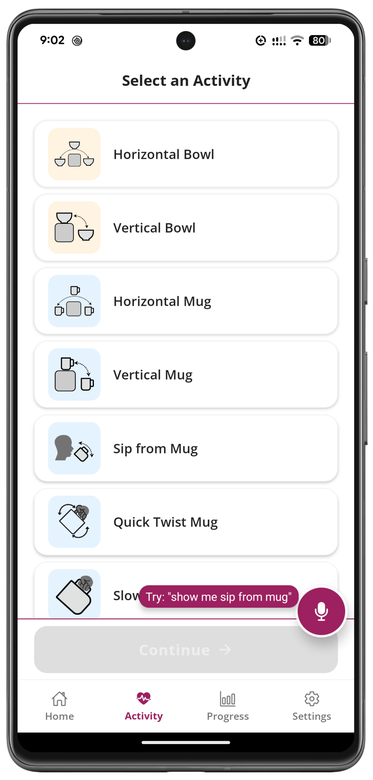
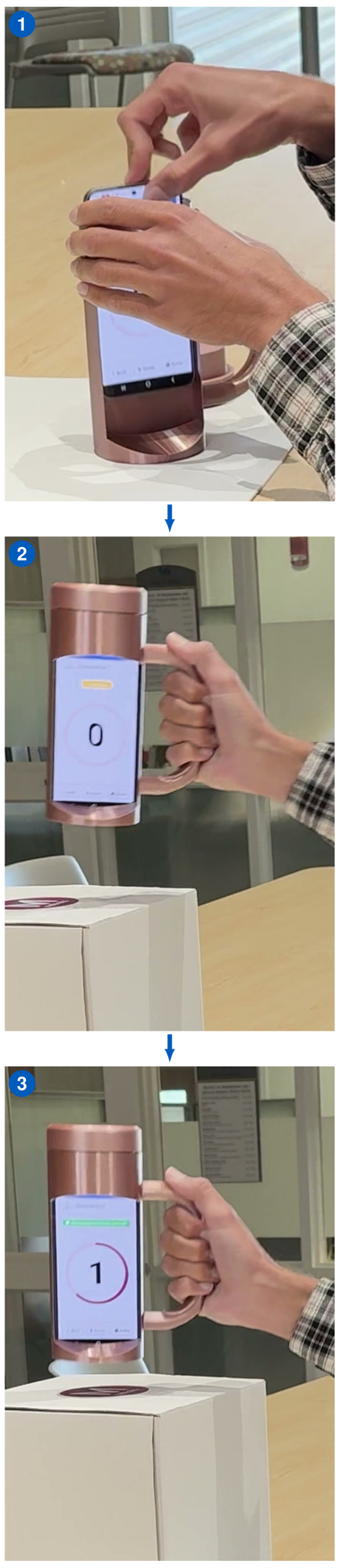


Figure B: *Choose from the activity library by voice or touch.*

A live repetition



Secure the phone in the mug.

Start sensing. The count shows 0.

Raise the mug. The count reaches 1.

Figure C: *Frames from the running app during a vertical-mug lift. Slow pour at the booth follows the same loop.*

How voice works on the phone

- After speech recognition, an ONNX router sorts commands, questions, and unknown utterances.
- A quantized 270M-parameter FunctionGemma model maps commands to 15 on-screen app tools.

Touch remains available throughout.

Try it without a network

Switch on airplane mode and do it again.

- Start practice with on-phone instructions.
- Move the mug for sensing, counting, and scoring.
- Follow visual cues, spoken cues, and pouring sounds.
- Review local progress without a network.

Sessions save locally and queue for upload. Models and media are pre-provisioned.

How it works on the phone

Sense → Detect → Score → Review

- At 50 Hz, activity-specific state machines use the accelerometer and gyroscope to check tilt, timing, and speed.
- The pouring detector requires a 600-ms upright baseline, a controlled tilt of at least 15° below 45°/s, and a return toward the start orientation.
- Smoothness comes from RMS linear jerk. Lower is smoother.
- The app compares each session with the previous attempt and weekly record.

Review your practice

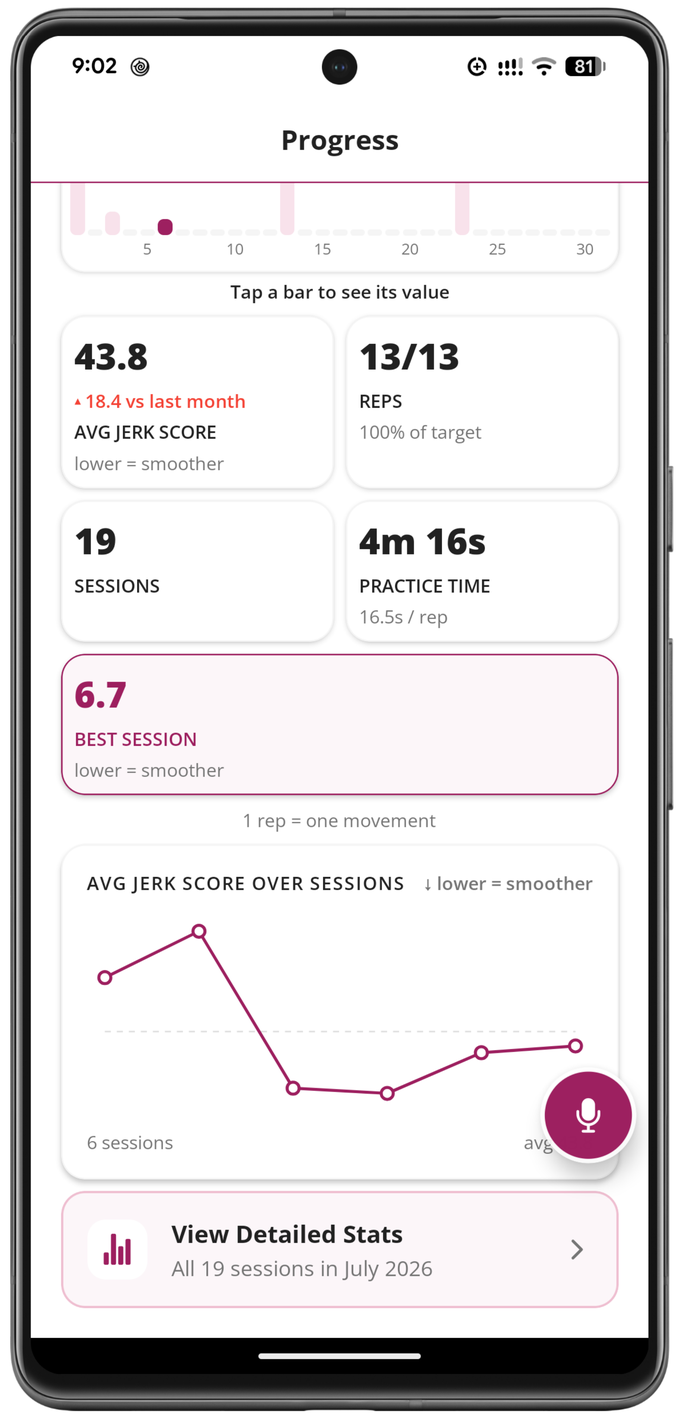


Figure D: *Practice counts, time, and smoothness scores. Illustrative values, not study outcomes.*

Prior studies of earlier versions

Earlier work tested measurement consistency and usability. A six-week in-home study followed 16 people with chronic stroke [1]. In a community study, 31 clinicians, survivors, and caregivers named accessibility and objective feedback as strengths. They identified personalization and fit with clinical care as priorities [2].

MobiCom '26 Demo. DOI: 10.1145/3795866.3848473.

[1] Langan et al., *JMIR mHealth and uHealth*, 2020. DOI: 10.2196/19582.

[2] Das et al., *Archives of Physical Medicine and Rehabilitation*, 2026. DOI: 10.1016/j.apmr.2026.02.218.

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